AYUSH

ML-Based Remaining Useful Life Prediction for Defence Equipment

Explainable predictive maintenance for equipment health and decision support

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1. Problem Statement

Degradation before failure

Equipment can gradually degrade before component failure, making health difficult to assess from isolated observations.

Operational impact

Unexpected failures can contribute to downtime, increased maintenance requirements and reduced equipment availability.

Opportunity for ML

Historical sensor measurements can reveal degradation patterns that may support earlier maintenance decisions.

Research question

Can machine-learning models estimate Remaining Useful Life (RUL) from historical sensor measurements while providing interpretable information for maintenance decision-making?

2. Proposed Solution

Sensor Data	Preprocessing	Feature Engineering	ML Models	RUL Prediction	Explainability	Maintenance Risk

Primary prediction

Estimate the number of remaining operating cycles before failure (RUL).

Decision support

Translate predicted RUL into prototype maintenance-risk information and explain the main contributing factors.

3. Dataset

NASA C-MAPSS — FD001

Publicly available turbofan-engine degradation simulation data used as a reproducible benchmark for prognostics and Remaining Useful Life prediction.

Data type

Multivariate time-series sensor observations across operating cycles.

100

Training trajectories

100

Test trajectories

1

Operating condition

1

Fault mode

Why this dataset?

It provides degradation trajectories with a known end-of-life point, enabling supervised RUL target construction and reproducible model comparison.

Source: NASA Prognostics Center of Excellence Data Repository

4. Machine Learning Methodology

RUL target

For each training trajectory, RUL is derived from the final failure cycle and the current operating cycle.

Feature engineering

Investigate current sensor values, cycle-to-cycle changes, rolling statistics and recent degradation trends.

Leakage control

Keep equipment trajectories separated during evaluation so future observations from the same equipment do not leak into training.

Models

Linear Regression
Baseline

Random Forest
Nonlinear ensemble

XGBoost
Gradient boosting

5. Evaluation & Explainability

MAE

Average absolute difference between predicted and actual RUL.

RMSE

Penalizes larger RUL prediction errors more strongly.

R^2

Measures how much variation in RUL is explained by the model.

Model	MAE	RMSE	R^2
Linear Regression	TBD	TBD	TBD
Random Forest	TBD	TBD	TBD
XGBoost	TBD	TBD	TBD

Explainability

Feature importance and SHAP-based analysis will be investigated to identify sensor characteristics contributing to individual RUL predictions.

Results will be added only after model training and validation.

6. Expected Outcome & Future Scope

Expected outcome

- RUL prediction from historical sensor data
- Comparison of multiple regression models
- Interpretable degradation-related features
- Prototype maintenance-risk information
- Reproducible ML workflow and documentation

Future scope

- Additional operating conditions and fault modes
- Real-world equipment sensor data
- Real-time health monitoring
- Edge/embedded model deployment
- Integration with maintenance-management systems
- Domain-expert validation

Project status: Proposal / Development in Progress

Project Repository | github.com/tisya-ahuja/AYUSH